

Table 1: Kuo Quadrupole R0500 (exact sphere surface $|\mathbf{r}| = 75 \text{ } \mu\text{m}$)

$$\text{Error} = \frac{1}{N} \sum_{i=1}^N \frac{|\mathbf{B}_{model} - \mathbf{B}_{FEM}|_i}{|\mathbf{B}_{FEM}|_i} \cdot 100$$

Kuo Quadrupole (Error=7.76%)		
Pole	$ \ell \text{ } [\mu\text{m}]$	Tip→Charge $[\mu\text{m}]$
P1	503.9	3.9
P2	503.9	3.9
P3	503.9	3.9
P4	503.9	3.9
Average	503.9	3.9
R_a	$1.36 \times 10^9 \text{ A/Wb}$	

Tip→Origin = 500 μm

Tip→Charge = $|\ell| - 500$

$N_c = 4$ (4 turns per pole), 4-fold symmetric quadrupole, sphere-surface fit (KDTree IDW interpolation onto $|r| = 75 \text{ } \mu\text{m}$) ($N = 800$ FEM nodes total).